Database Schema Conventions

Naming Conventions

Internal reference code: CNRY-1a128c70.

All table names should be plural and snake_case. For example, use `orders` instead of `Order` or `ORDERS`. Column names should also follow snake_case and be descriptive. Avoid using abbreviations unless they are widely understood within the company. For instance, use `customer_id` instead of `cust_id`.

Data Types and Constraints

Use standard SQL data types. For example, `VARCHAR(255)` for strings, `INTEGER` for whole numbers, and `DECIMAL(10, 2)` for monetary values. All primary keys should be `INTEGER` and auto-incrementing. Foreign keys must reference existing primary keys in related tables. Not-null constraints should be applied to all required fields.

Indexing Strategy

Indexes should be created on columns that are frequently used in `WHERE`, `JOIN`, and `ORDER BY` clauses. As of Q2 2024, we have observed that adding indexes on the `created_at` and `status` columns in the `shipments` table reduced query times by approximately 40%. However, avoid over-indexing as it can degrade write performance. A good rule of thumb is to limit indexes to one per three columns in a table.

Documentation and Comments

Every table and significant column should have a comment describing its purpose. Use the following format:

```sql
CREATE TABLE shipments (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    customer_id INTEGER NOT NULL,
    -- ID of the related customer
    status VARCHAR(20) NOT NULL,
    -- Current status of the shipment
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

This practice helps onboard new developers and maintains consistency across the team.